

ZEYU ZHAO

[Google Scholar](#) [LinkedIn](#) [Personal Website](#)

London, England

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RESEARCH INTERESTS

Generative AI, End-to-End (E2E) Automatic Speech Recognition (ASR), Text-to-Speech (TTS), Speech Large Language Models (LLMs)

SKILLS

- Languages, Libraries, & Tools: Python, C++, Shell, PyTorch, k2, Git
- Machine Learning: Transformer, CTC, Seq2Seq
- Research: ASR, E2E ASR
- Software Engineering: Open-source Development, Modular Framework Design

EDUCATION

University of Edinburgh

Sept. 2020 - Jul. 2025

PhD, supervised by [Prof. Peter Bell](#)
End-to-End Speech Recognition
[Centre for Speech Technology Research](#)
[School of Informatics](#)

Tsinghua University

Sept. 2017 - Aug. 2020

Master of Science, supervised by [Dr. Wei-Qiang Zhang](#)
End-to-End Keyword Search and Speech Recognition for Low Resource Languages
[Speech and Audio Technology Lab](#)
Department of Electronic Engineering

CGPA: 3.6/4.0

Beijing Institute of Technology

Sept. 2013 - Jun. 2017

Bachelor of Science
Department of Electronic Engineering

CGPA: 90/100

EXPERIENCE

Machine Learning Engineer

Jan. 2026 - present

Speech Technology
Speechmatics
Cambridge & London, England

Research Officer (postdoc)

Sept. 2024 - Dec. 2025

Generative and Interactive AI
EVE project, funded by EPSRC, held by [Prof. Matt Jones](#)
Department of Computer Science, Swansea University
Swansea, Wales

Research Associate (part-time)

Mar. 2024 - Aug. 2024

Post-training and evaluating Large Language Models (LLMs) for finance application
Main contributor and code writer

School of Informatics, University of Edinburgh
Edinburgh, Scotland

Teaching Support (part-time)

Jan. 2022 - Mar. 2024

Preparing teaching materials and marking courseworks
School of Informatics, University of Edinburgh
Edinburgh, Scotland

OPEN-SOURCE CONTRIBUTION

SpeechBrain k2 integration

A popular open-source speech processing toolkit on [Github](#)

Contributor

Contributed to integration of k2-based training and decoding into SpeechBrain's ASR pipeline.

Ben Nevis

An End-to-End ASR framework based on PyTorch and [k2](#), a Differentiable Weighted Finite-State Transducer library

Creator

Please see my personal [Github](#) for more information.

SELECTED PUBLICATIONS

For the full list, please see my [Google Scholar](#).

Z. Zhao and P. Bell, *Regarding the Existence of the Internal Language Model in CTC-Based E2E ASR*, ICASSP 2025

Z. Zhao and P. Bell, *Advancing CTC Models for Better Speech Alignment: A Topological Approach*, SLT 2024

Z. Zhao, P. Chen and P. Bell, *Regarding Topology and Adaptability in Differentiable WFST-Based E2E ASR*, ICASSP 2024 XAI-SA Workshop

Z. Zhao, P. Bell and O. Klejch, *Exploring Dominant Paths in CTC-Like ASR Models: Unraveling the Effectiveness of Viterbi Decoding*, ICASSP 2024 XAI-SA Workshop

Z. Zhao and P. Bell, *Regarding Topology and Variant Frame Rates for Differentiable WFST-based End-to-End ASR*, Interspeech 2023

Y. Li[†], **Z. Zhao**[†], O. Klejch, P. Bell, and C. Lai, *A Word-Level Investigation of the Mutual Impact of Speech and Emotion Recognition*, Interspeech 2023

Z. Zhao and P. Bell, *Investigating Sequence-Level Normalisation For CTC-Like End-to-End ASR*, ICASSP 2022

Z. Zhao and W. Q. Zhang, *End-to-End Keyword Search Based on Attention and Energy Scorer for Low Resource Languages*, Interspeech 2020

Z. Zhao and W. Q. Zhang, *End-to-End Keyword Search System Based on Attention Mechanism and*

[†]Equal contribution.

AWARDS & SCHOLARSHIPS

Student Research Award

2019

Research Center for Media Big Data Cognitive Computing, Department of Electronic Engineering
Tsinghua University, Beijing, China

National Scholarship

2014 - 2016

Beijing Institute of Technology, Beijing, China
For top students only

First Undergraduate Scholarship

2013 - 2017

Beijing Institute of Technology, Beijing, China
For top students only

Third Prize, Beijing Electronic Design Competition

2016

Honorable Mentioned, Mathematical Contest in Modelling

2015

First Prize in National Physics Competition

2014